

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. BPT-503: Quantum Mechanics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 4 = 16]

- (a) An X-ray photon of wavelength $1.01 \text{ \AA}$ is scattered off a stationary electron at ninety degrees to the direction of incidence. Find the wavelength of the scattered photon and the recoil angle of the electron.
- (b) Given four operators A , B , C , and D , find which of these fulfill the condition of being a linear operator. Given:

$$Af(x) = f(x) + b; \quad Bg(x) = [g(x)]^2; \quad Ch(x) = ch(x); \quad Dy(x) = \frac{dy(x)}{dx}$$

where $f(x), g(x), h(x), y(x)$ are functions and b, c are constants.

- (c) Find the result of the following commutator brackets:
- (i) $[L^2, L_x]$
- (ii) $[x, L_y]$
- (d) What do you understand by the momentum operator? Does it commute with the Hamiltonian operator for a free particle? Explain briefly.
- (e) Find the de Broglie wavelength of: (i) a proton of kinetic energy 1 MeV, and (ii) an electron of kinetic energy 1 MeV.

Question 2.

- (a) A particle of energy E is incident on a rectangular potential barrier of width a and height U :

$$V(x) = \begin{cases} U, & 0 < x < a \\ 0, & x < 0 \text{ and } x > a \end{cases}$$

If the energy $E < U$, solve the Schrödinger equation with appropriate boundary conditions to get the reflection coefficient R and transmission coefficient T . [7.5]

- (b) How do you explain the presence of the particle in the classically forbidden region? Give two examples from physics where you can observe this tunneling phenomenon. [6.5]

Question 3.

- (a) The potential energy function of a linear harmonic oscillator (LHO) is $V(x) = \frac{1}{2}kx^2$. Is this a problem where defining the parity operator is useful? Prove that the eigenvalues of the parity operator are good quantum numbers. [4]
- (b) Find the ground state wave function for the LHO by using the lowering operator a_- :

$$a_- = \frac{1}{\sqrt{2\hbar m\omega}}(m\omega x + ip)$$

Normalize the wave function obtained. [6]

- (c) Find the expectation value of the position operator x for the ground state. [4]

Question 4.

- (a) Write the time-independent Schrödinger equation for the Hydrogen atom in spherical polar coordinates and carry out the separation of variables. Solve the radial equation and find the expression for the energy eigenvalues. Explain the significance of the quantum numbers n , l , and m . [10.5]
- (b) Normalize the ground state wave function for the Hydrogen atom:

$$\psi(r, \theta, \phi) = Ne^{-r/a}$$

where a is the Bohr radius. [3.5]

OR

- (a) Given the amplitude function in k -space:

$$A(k) = \begin{cases} A_0, & -K < k < K \\ 0, & \text{otherwise} \end{cases}$$

where A_0 is a constant. Find the Fourier transform of $A(k)$ defined as:

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} A(k) dk$$

Sketch the wavepacket $\psi(x)$ so obtained. If Δk is the width of $A(k)$ and Δx is the width of the wavepacket, then find and interpret the product $\Delta k \Delta x$. [7]

- (b) What do you understand by commutativity and compatibility of operators? Explain briefly. [2]
- (c) Derive the uncertainty relation between two non-commuting operators A and B . Deduce the minimum uncertainty product. [5]

Question 5.

- (a) What do you understand by wave-particle duality of matter? Name two experiments that illustrate this duality. [3]

- (b) Describe the experiment first performed by Davisson and Germer to show electron diffraction. [5]
- (c) If an electron is perfectly confined to move in a 1D box of length L (potential $V(x) = 0$ for $0 < x < L$, and $V(x) = \infty$ otherwise), find the set of possible energy values the electron may have. Also find the ground state and first excited state wave functions for the electron in the box. Normalize these wave functions. [6]
- (d) Given $L = 1 \text{ \AA}$, find the lowest possible energy the electron might have in eV units. [2]

OR

- (a) Use the angular momentum ladder operators L_+ and L_- and the commutation relations to find the eigenvalues of the operators L^2 and L_z . [7]
- (b) Given the normalized wave function for the 1s state of the Hydrogen atom:

$$\psi_{100}(r) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$$

where a is the Bohr radius, find:

- (i) the most probable distance of the electron from the origin. [3.5]
- (ii) the expectation value of the operator r . [3.5]